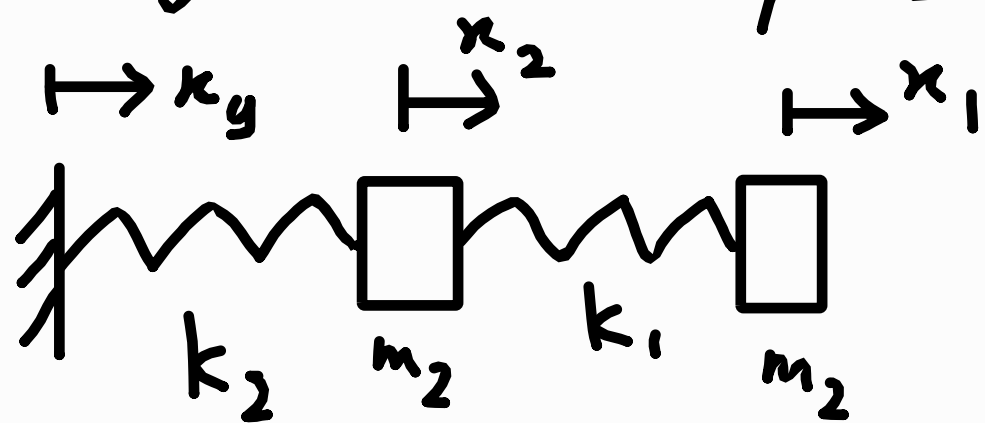
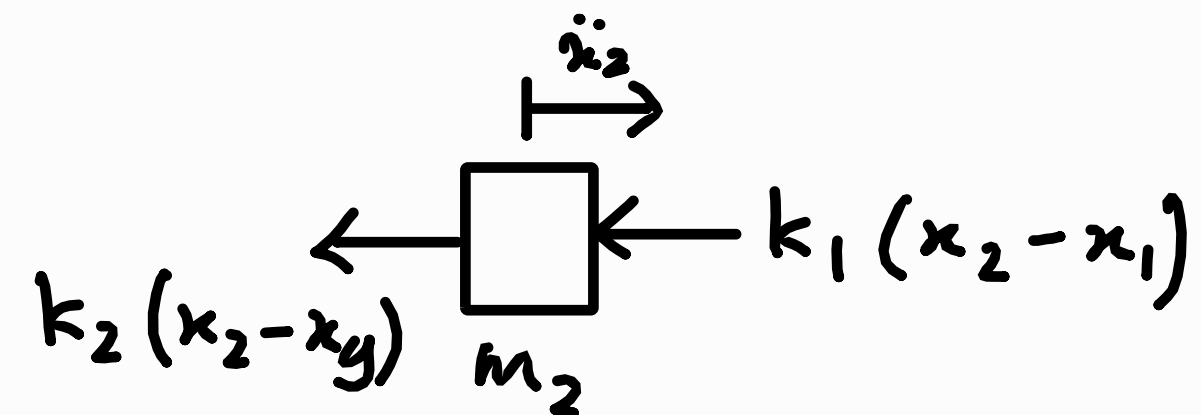


1) Equivalent system



Free body diagram:



Assuming $x_2 > x_1$, $x_2 > x_g$

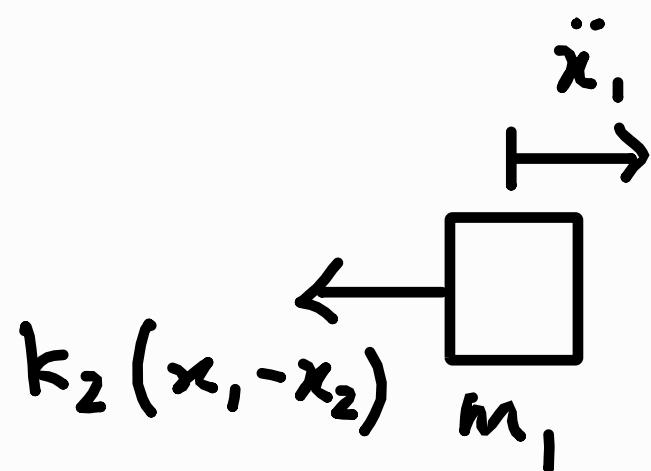
By Newton's 2nd law,

$$m_2 \ddot{x}_2 = -[k_1(x_2 - x_1) + k_2(x_2 - x_g)]$$

$$m_2 \ddot{x}_2 + k_1(x_2 - x_1) + k_2(x_2 - x_g) = 0$$

$$m_2 \ddot{x}_2 + k_1 x_2 - k_1 x_1 + k_2 x_2 = k_2 x_g$$

$$m_2 \ddot{x}_2 - k_1 x_1 + (k_1 + k_2) x_2 = k_2 x_g$$



Assuming $x_1 > x_2$

$$m_1 \ddot{x}_1 = -k_1(x_1 - x_2)$$

$$m_1 \ddot{x}_1 + k_1(x_1 - x_2) = 0$$

$$m_1 \ddot{x}_1 + k_1 x_1 - k_1 x_2 = 0$$

In matrix form:

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 + k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ k_2 x_g \end{Bmatrix} \sin \omega t$$

Assuming the steady state response is of the form:

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \sin \omega t, \quad \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} = -\omega^2 \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \sin \omega t$$

5) Substituting the equations above:

$$-w^2 \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} \cancel{\sin wt} + \begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1+k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} \cancel{\sin wt} \\ = \begin{Bmatrix} 0 \\ k_2 x_g \end{Bmatrix} \cancel{\sin wt}$$

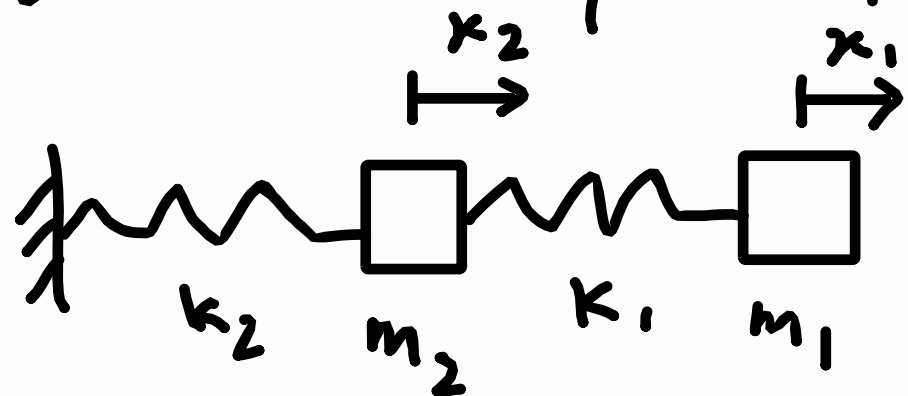
$$\begin{bmatrix} -w^2 m_1 + k_1 & -k_1 \\ -k_1 & -w^2 m_2 + k_1 + k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ k_2 x_g \end{Bmatrix}$$

Solving for the amplitudes using Cramer's rule:

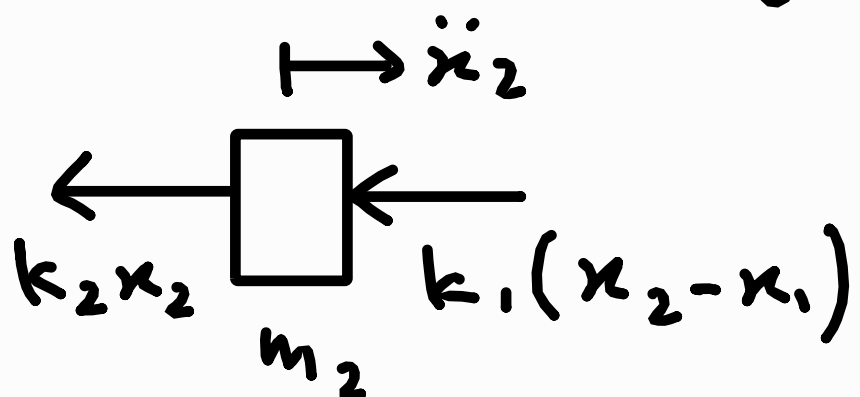
$$X_1 = \frac{\begin{vmatrix} 0 & -k_1 \\ k_2 x_g & -w^2 m_2 + k_1 + k_2 \end{vmatrix}}{\begin{vmatrix} -w^2 m_1 + k_1 & -k_1 \\ -k_1 & -w^2 m_2 + k_1 + k_2 \end{vmatrix}} \\ = \frac{k_1 k_2 x_g}{(-w^2 m_1 + k_1)(-w^2 m_2 + k_1 + k_2) - k_1^2}$$

$$X_2 = \frac{\begin{vmatrix} -w^2 m_1 + k_1 & 0 \\ -k_1 & k_2 x_g \end{vmatrix}}{\begin{vmatrix} -w^2 m_1 + k_1 & -k_1 \\ -k_1 & -w^2 m_2 + k_1 + k_2 \end{vmatrix}} \\ = \frac{(-w^2 m_1 + k_1) k_2 x_g}{(-w^2 m_1 + k_1)(-w^2 m_2 + k_1 + k_2) - k_1^2}$$

2) Equivalent system:



Free body diagram:



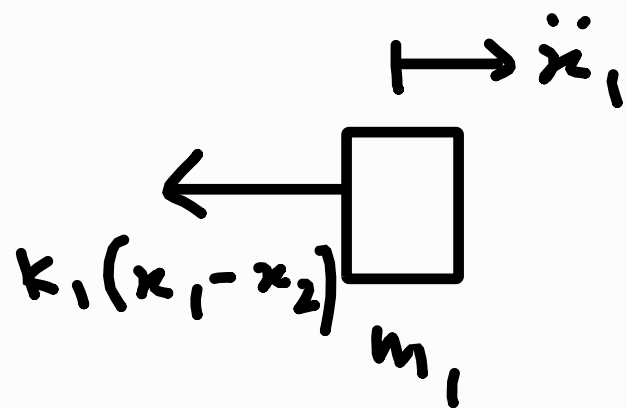
Assuming $x_2 > x_1$

By Newton's 2nd law,

$$m_2 \ddot{x}_2 = -(k_2 x_2 + k_1 (x_2 - x_1))$$

$$m_2 \ddot{x}_2 + k_2 x_2 + k_1 (x_2 - x_1) = 0$$

$$m_2 \ddot{x}_2 + (k_2 + k_1) x_2 - k_1 x_1 = 0$$



Assuming $x_1 > x_2$

$$m_1 \ddot{x}_1 = -k_1 (x_1 - x_2)$$

$$m_1 \ddot{x}_1 + k_1 x_1 - k_1 x_2 = 0$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 + k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Assuming the steady-state solution is of the form:

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \sin \omega t$$

$$\begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} = -\omega^2 \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \sin \omega t$$

2) Substituting the above equations:

$$-\omega^2 \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} \cancel{\sin \omega t} + \begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 + k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} \cancel{\sin \omega t} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} -\omega^2 m_1 + k_1 & -k_1 \\ -k_1 & -\omega^2 m_2 + k_1 + k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (1)$$

For a non-trivial solution, the determinant must be 0:

$$\begin{vmatrix} -\omega^2 m_1 + k_1 & -k_1 \\ -k_1 & -\omega^2 m_2 + k_1 + k_2 \end{vmatrix} = 0$$

$$k_1^2 - (-\omega^2 m_1 + k_1)(-\omega^2 m_2 + k_1 + k_2) = 0$$

Substituting $k_1 = k, k_2 = 2k, m_1 = m, m_2 = 2m$

$$k^2 - (-\omega^2 m + k)(-\omega^2 (2m) + 3k) = 0$$

$$k^2 - (2\omega^4 m^2 - 3k\omega^2 m - 2k\omega^2 m + 3k^2) = 0$$

$$2m^2\omega^4 - 5km\omega^2 + 2k^2 = 0$$

$$\omega^2 = \frac{5km \pm \sqrt{(-5km)^2 - 4(2m^2)(2k^2)}}{2(2m^2)}$$

$$= \frac{5k}{4m} \pm \frac{1}{4m^2} \sqrt{25k^2 m^2 - 16k^2 m^2}$$

$$= \frac{5k}{4m} \pm \frac{3k}{4m}$$

$$2) \quad \omega^2 = \frac{5k}{4m} \pm \frac{3k}{4m}$$

$$\omega^2 = \frac{8k}{4m}$$

$$\text{or} \quad \omega^2 = \frac{2k}{4m}$$

$$\omega = \sqrt{\frac{2k}{m}}$$

$$\omega = \sqrt{\frac{k}{2m}}$$

Mode shapes:

Using the 1st equation of (1):

$$(-\omega^2 m_1 + k_1) X_1 - k_1 X_2 = 0$$

$$(-\omega^2 m + k) X_1 = k X_2$$

$$\frac{X_2}{X_1} = \frac{-\omega^2 m + k}{k}$$

$$\text{When } \omega = \sqrt{\frac{2k}{m}},$$

$$\frac{X_2}{X_1} = \frac{-\frac{2k}{\cancel{m}} + k}{k}$$

$$= -1$$

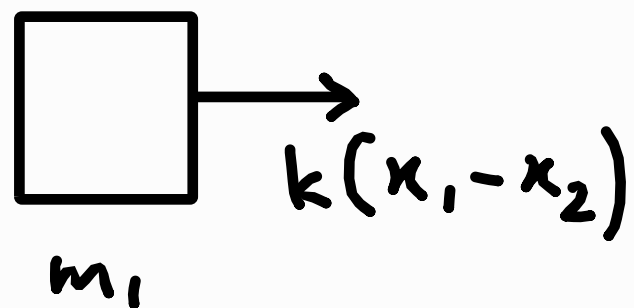
$$\text{When } \omega = \sqrt{\frac{k}{2m}},$$

$$\frac{X_2}{X_1} = \frac{-\frac{k}{2\cancel{m}} + k}{k}$$

$$= \frac{1}{2}$$

3) Free body diagrams:

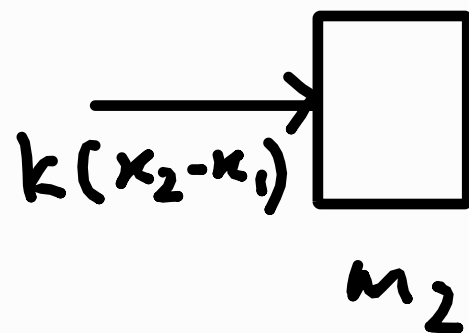
$$x_1, \ddot{x}_1 \leftarrow$$



$$m_1 \ddot{x}_1 = -k(x_1 - x_2)$$

$$m_1 \ddot{x}_1 + kx_1 - kx_2 = 0$$

$$x_2, \ddot{x}_2 \leftarrow$$



$$m_2 \ddot{x}_2 = -k(x_2 - x_1)$$

$$m_2 \ddot{x}_2 - kx_1 + kx_2 = 0$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Assuming the steady-state response is of the form:

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \sin \omega t \quad \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} = -\omega^2 \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \sin \omega t$$

Substituting the above:

$$-\omega^2 \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \cancel{\sin \omega t} + \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \cancel{\sin \omega t} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} k - \omega^2 m_1 & -k \\ -k & k - \omega^2 m_2 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

3) For a non-trivial solution, the determinant must be 0:

$$\begin{vmatrix} k - \omega^2 m_1 & -k \\ -k & k - \omega^2 m_2 \end{vmatrix} = 0$$

$$(k - \omega^2 m_1)(k - \omega^2 m_2) - k^2 = 0$$

$$\cancel{k^2} - k\omega^2 m_2 - k\omega^2 m_1 + \omega^4 m_1 m_2 - \cancel{k^2} = 0$$

$$m_1 m_2 \omega^4 - k(m_1 + m_2)\omega^2 = 0$$

$$\omega^2 (m_1 m_2 \omega^2 - k(m_1 + m_2)) = 0$$

$$\omega^2 = 0 \quad \text{or} \quad m_1 m_2 \omega^2 = k(m_1 + m_2)$$

$$\omega = 0$$

$$\omega = \sqrt{\frac{k(m_1 + m_2)}{m_1 m_2}}$$

$$\therefore f_1 = 0 \quad \text{or}$$

$$f = \frac{1}{2\pi} \omega = \frac{1}{2\pi} \sqrt{\frac{k(m_1 + m_2)}{m_1 m_2}}$$

$$= \frac{1}{2\pi} \sqrt{\frac{175 \times 10^3 (1750 + 3800)}{1750 \times 3800}}$$

$$= 1.923423088 \text{ Hz}$$

$$\approx 1.92 \text{ Hz}$$

3) Mode shapes:

$$k - \omega^2 m_1 X_1 - k X_2 = 0$$

$$\frac{X_2}{X_1} = \frac{k - \omega^2 m_1}{k}$$

When $\omega = 0$,

$$\begin{aligned} \frac{X_2}{X_1} &= \frac{k - 0}{k} \\ &= 1 \end{aligned}$$

When $\omega = 2\pi(1.92)$,

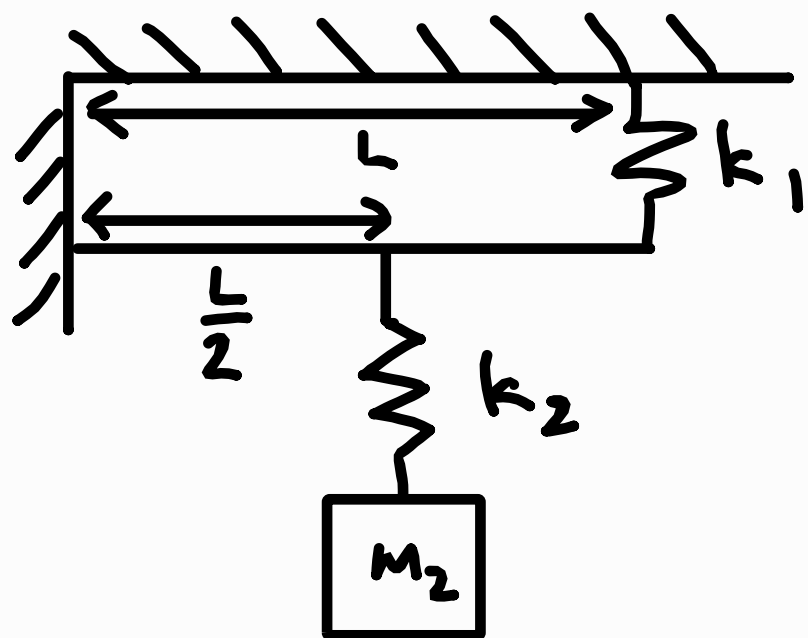
$$\frac{X_2}{X_1} = \frac{175 \times 10^3 - (2\pi(1.92))^2 \times 1750}{175 \times 10^3}$$

$$= -0.4605263158$$

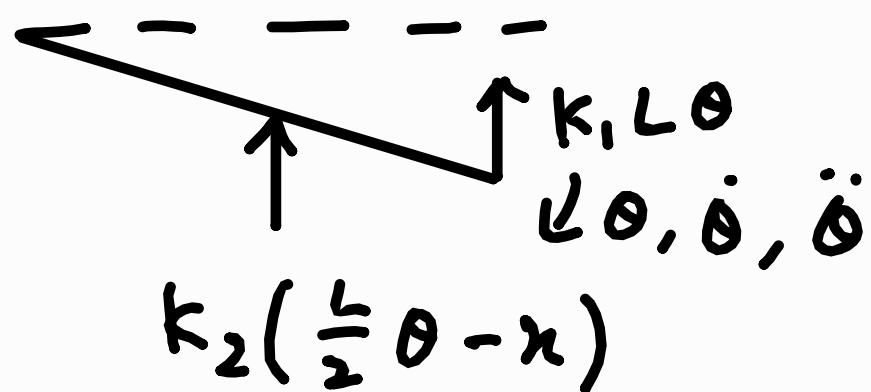
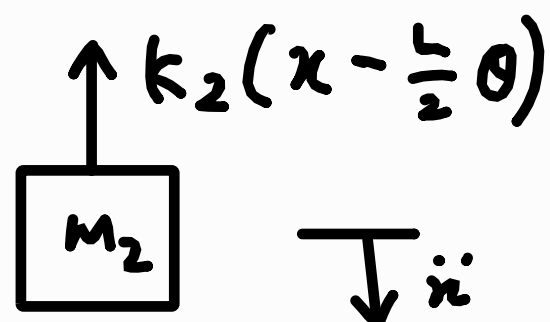
$$\approx -0.46$$

$$\therefore \begin{Bmatrix} \phi_{12} \\ \phi_{22} \end{Bmatrix} = \begin{Bmatrix} 1 \\ -0.46 \end{Bmatrix}$$

4)



Free body diagram:

Assuming $\frac{L}{2}\theta > x$ Assuming $x > \frac{L}{2}\theta$

For the lever:

$$J_A \ddot{\theta} = \sum M_A$$

$$J_A \ddot{\theta} = -k_1 (L\theta - 0)L - k_2 \left(\frac{L}{2}\theta - x\right) \frac{L}{2}$$

$$J_A \ddot{\theta} + \left(k_1 L^2 + \frac{k_2 L^2}{4}\right) \theta - \frac{k_2 L x}{2} = 0$$

For the mass:

$$m_2 \ddot{x} = -k_2 \left(x - \frac{L}{2}\theta\right)$$

$$m_2 \ddot{x} - \frac{k_2 L}{2} \theta + k_2 x = 0$$

In matrix form:

$$\begin{bmatrix} J_0 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{\theta} \\ \ddot{x} \end{Bmatrix} + \begin{bmatrix} k_1 L^2 + \frac{k_2 L^2}{4} & -\frac{k_2 L}{2} \\ -\frac{k_2 L}{2} & k_2 \end{bmatrix} \begin{Bmatrix} \theta \\ x \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

4) Assuming the steady-state solution is of the form:

$$\begin{Bmatrix} \theta \\ x \end{Bmatrix} = \begin{Bmatrix} \Theta \\ X \end{Bmatrix} \sin \omega t \quad \begin{Bmatrix} \ddot{\theta} \\ \ddot{x} \end{Bmatrix} = -\omega^2 \begin{Bmatrix} \Theta \\ X \end{Bmatrix} \sin \omega t$$

Substituting the above:

$$-\omega^2 \begin{bmatrix} J_0 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \Theta \\ X \end{Bmatrix} \sin \omega t + \begin{bmatrix} k_1 L^2 + \frac{k_2 L^2}{4} & -\frac{k_2 L}{2} \\ -\frac{k_2 L}{2} & k_2 \end{bmatrix} \begin{Bmatrix} \Theta \\ X \end{Bmatrix} \sin \omega t = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} -\omega^2 J_0 + k_1 L^2 + \frac{k_2 L^2}{4} & -\frac{k_2 L}{2} \\ -\frac{k_2 L}{2} & -\omega^2 m_2 + k_2 \end{bmatrix} \begin{Bmatrix} \Theta \\ X \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

For a non-trivial solution, the determinant must be 0:

$$\begin{vmatrix} -\omega^2 J_0 + k_1 L^2 + \frac{k_2 L^2}{4} & -\frac{k_2 L^2}{4} \\ -\frac{k_2 L}{2} & -\omega^2 m_2 + k_2 \end{vmatrix} = 0$$

$$\left(-\omega^2 J_0 + k_1 L^2 + \frac{k_2 L^2}{4}\right) \left(-\omega^2 m_2 + k_2\right) + \frac{k_2^2 L^2}{4} = 0$$

$$\omega^4 J_0 m_2 - \omega^2 m_2 k_1 L^2 - \omega^2 m_2 \frac{k_2 L^2}{4} - \omega^2 k_2 J_0 + k_1 k_2 L^2 - \frac{k_2^2 L^2}{4} + \frac{k_2^2 L^2}{4} = 0$$

$$J_0 m_2 \omega^4 - \left(m_2 k_1 L^2 + \frac{m_2 k_2 L^2}{4} + k_2 J_0\right) \omega^2 + k_1 k_2 L^2 = 0$$

4) Substituting $k_1 = 10 \times 10^3 \text{ N m}^{-1}$
 $k_2 = 20 \times 10^3 \text{ N m}^{-1}$
 $m_1 = 5 \text{ kg}$
 $m_2 = 3 \text{ kg}$

$$L = 0.3 \text{ m}$$

$$J_0 = \frac{1}{3} m L^2 = \frac{1}{3} (5) (0.3)^2$$

$$= 0.15 \text{ kg m}^2$$

$$0.15 (3) \omega^4 - (3 \times 10 \times 10^3 (0.3)^2 + \frac{3 \times 20 \times 10^3 (0.3)^2}{4} + 20 \times 10^3 \times 0.15) \omega^2$$

$$- 20 \times 10^3 \times 10 \times 10^3 (0.3)^2 = 0$$

$$0.45 \omega^4 - 7050 \omega^2 + 18000000 = 0$$

Solving:

$$\omega^2 = 12455.14154$$

$$\text{or } \omega^2 = 3211.525125$$

$$\omega = 111.6026054$$

$$\text{or } \omega = 56.6703196$$

$$\approx 111.6 \text{ rads}^{-1}$$

$$\approx 56.67 \text{ rads}^{-1}$$

The first natural frequency is the lowest,

$$\therefore f_1 = \frac{1}{2\pi} (56.67)$$

$$= 0.01936149$$

$$\approx 0.02 \text{ Hz}$$